ARSALAN FIROZ

Backend Software Engineer
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PROFESSIONAL SUMMARY

Backend Software Engineer with 4.5+ years of experience building high-throughput, event-driven microservices for consumer fintech platforms. Production experience in payment gateway integration, promotions and rewards engines, fraud prevention, and database performance tuning using Node.js, TypeScript, MySQL, MongoDB, Redis, Kafka, AWS, Docker, and Kubernetes. Built services supporting 50+ domains across 10+ countries and millions of daily transactions. Core areas include system design, distributed systems, and data structures and algorithms. Hands-on with GenAI: LLM applications, RAG pipelines, AI agents, and LLM-assisted development.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL

Backend and APIs: Node.js, Express.js, FastAPI, REST APIs, GraphQL, gRPC, WebSocket, Microservices, API Design, JWT, OAuth 2.0, Rate Limiting

Distributed Systems and Messaging: System Design, Event-Driven Architecture, Apache Kafka, Message Queues, Message Brokers, High Availability, Scalability, Load Balancing, Caching, Design Patterns

Databases: MySQL, PostgreSQL, MongoDB, Redis, DuckDB, SQL and NoSQL, Query Optimization, Database Performance Tuning, Schema Design

Cloud and DevOps: AWS, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Linux, Git

GenAI and LLM: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, LangGraph, Vector Databases, Embeddings, Semantic Search, AI Agents, Prompt Engineering, Ollama, OpenAI, Gemini, Anthropic APIs

Practices: Data Structures and Algorithms, OOP, Test-Driven Development, Code Review, Agile, SDLC, Jira, Technical Documentation

PROFESSIONAL EXPERIENCE

Software Engineer, FPLE SDN BHD | Bengaluru, India | May 2024 - Present

- Designed and scaled backend microservices supporting 50+ domains across 10+ countries, processing millions of daily transactions on consumer-facing platforms.
- Built promotions and rewards services (cashback, instant rebate, loyalty programs) that contributed to a 60% increase in new-user registrations and a 75% improvement in participation.
- Contributed to payment gateway integrations handling millions of transactions, improving payment success rates by 30%.
- Developed high-performance REST APIs and financial-service integrations in Node.js and TypeScript for production-critical transaction workflows.
- Implemented fraud-prevention controls (OTP verification, rate limiting, IP whitelisting), reducing fraudulent transaction attempts by approximately 95%.
- Optimized MySQL and MongoDB queries through production log analysis and indexing, reducing transaction failures by 50% and improving database response times by 35%.
- Managed production deployments and CI/CD pipelines using Jenkins and Docker with minimal downtime.
- Contributed to system design discussions for distributed backend systems, designing scalable solutions for growing traffic and transaction volumes.
- Mentored junior engineers through code reviews and knowledge-sharing sessions, improving code quality and delivery velocity.
- Used LLMs for code generation, debugging, test writing, and internal automation to speed up development.

Software Engineer, Cognizant | Bengaluru, India | Jan 2022 - Apr 2024

Promoted from Junior Software Engineer (Jan 2022 - Feb 2023) to Software Engineer (Mar 2023 - Apr 2024).

- Completed a 6-month full-stack training program, building a trainee management system end to end as part of a team, then moved into client project delivery.
- Developed and maintained backend modules and REST APIs in collaboration with senior engineers.
- Wrote optimized SQL queries and contributed to database schema design to improve data retrieval efficiency.
- Participated in code reviews, unit testing, and debugging following SDLC and Agile practices.
- Collaborated using Git and Jira across delivery workflows.

PROJECTS

Enterprise RAG-Based AI Knowledge Worker and Business Intelligence Engine | Python, FastAPI, Ollama, DuckDB

- Built an offline-first RAG application that answers questions across unstructured documents (PDF, DOCX, Markdown, CSV) and structured business data.
- Implemented hybrid retrieval combining dense embeddings (BAAI/bge-small-en-v1.5, FastEmbed, ONNX Runtime) with BM25 keyword search, using recursive paragraph-aware chunking with overlap.
- Added heuristic query routing between document RAG and DuckDB SQL analytics, with citations and anti-hallucination guardrails (source titles, chunk IDs, relevance scores).
- Built asynchronous FastAPI REST APIs and a web UI; integrated Ollama for local LLM inference with no mandatory external API calls.

BrandSynth AI: Business Intelligence and B2B Brochure Generator | Python, Flask, LLM APIs

- Built an application that converts corporate website URLs into structured company insights and B2B sales brochures, exported as Markdown, HTML, PDF, and JSON.
- Scraped and cleaned web content with Requests and BeautifulSoup4; classified relevant pages using zero-shot classification and LLM reasoning.
- Integrated Gemini, OpenAI, Groq, and Anthropic APIs with structured JSON outputs and provider fallback.
- Wrote automated unit and integration tests with mocked network requests.

Ticket Booking System: Event-Driven Microservices | Node.js, Docker, Kubernetes

- Designed independently deployable Auth, Orders, Payments, and Moderation microservices communicating asynchronously through an event bus.
- Solved distributed data consistency using event synchronization patterns.
- Deployed on Kubernetes with Ingress, load balancing, and service-to-service communication; secured with JWT authentication.
- Built CI/CD pipelines for automated testing and production-ready deployments.

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate (Oct 2021)
- Oracle Cloud Infrastructure 2021 Certified Architect Associate (Oct 2021)
- AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents, Udemy (Oct 2026)
- Claude Code in Action, Anthropic (Mar 2026)
- Hands On AI: Build Your Own GPTs, LinkedIn (Oct 2025)
- Generative AI Fundamentals, Databricks (Jul 2025)
- ChatGPT Prompt Engineering for Developers, DeepLearning.AI (Jul 2025)
- HackerRank: Software Engineer, SQL (Intermediate), REST API (Intermediate), JavaScript (Intermediate) (Apr 2025)
- Postman API Fundamentals Student Expert (Jan 2025)
- GitHub Copilot Beginner to Pro, Udemy (Oct 2024)
- LFD103: A Beginner's Guide to Linux Kernel Development, The Linux Foundation (Dec 2022)

EDUCATION

Bachelor of Technology (B.Tech), Computer Science and Engineering | University of Allahabad, J.K. Institute of Applied Physics and Technology

ISC (Class XII), PCM with Computer Science | ADLS Sunshine School

LANGUAGES

English (Full professional proficiency), Hindi (Full professional proficiency)